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* ===========================================================
* Telecom Staff Scheduling (GAMS) - all 3 notebook parts
* ===========================================================

Sets
    h   "hours"     /h0*h23/
    j   "companies" /c1*c2/
    peak(h) "hours 7..17 (inclusive)"
    off(h)  "hours 0..6 and 18..23"
    Q(h,h)  "coverage mapping: for each demand hour, staffed start hours in 6-hour window";

Alias (h,i,k);

* Subsets for objective weights (ord(h0)=1)
peak(h) = yes$(ord(h) >= 8 and ord(h) <= 18);
off(h)  = yes$(ord(h) <= 7 or ord(h) >= 19);

Parameters
    cost(h)        "cost per operator starting at hour"
    demand(h,j)    "demand by hour and company"
    capHour(h)     "RHS for per-hour starting cap"
    capTotal(h)    "RHS for total active cap per demand hour"
    minCov(h)      "minimum coverage per demand hour"
    maxShort(h,j)  "max shortage per hour/company"
    totShort(j)    "total shortage cap by company";

* --- Default RHS values (so we can vary them later in part 3)
capHour(h) = 100;
capTotal(h)= 420;
minCov(h)  = 80;
maxShort(h,j) = 10;
totShort('c1') = 40;
totShort('c2') = 30;

* Costs (hour 0..23)
cost('h0')  = 189;  cost('h1')  = 189;  cost('h2')  = 189;  cost('h3')  = 180;
cost('h4')  = 171;  cost('h5')  = 162;  cost('h6')  = 153;  cost('h7')  = 144;
cost('h8')  = 135;  cost('h9')  = 126;  cost('h10') = 126;  cost('h11') = 135;
cost('h12') = 144;  cost('h13') = 153;  cost('h14') = 162;  cost('h15') = 171;
cost('h16') = 180;  cost('h17') = 189;  cost('h18') = 189;  cost('h19') = 189;
cost('h20') = 189;  cost('h21') = 189;  cost('h22') = 189;  cost('h23') = 189;

* Demand (hour, company)
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```
* Company 1
demand('h0','c1')  = 120; demand('h1','c1')  = 95;  demand('h2','c1')  = 90;  demand('h3','c1')  = 50;
demand('h4','c1')  = 30;  demand('h5','c1')  = 30;  demand('h6','c1')  = 60;  demand('h7','c1')  = 110;
demand('h8','c1')  = 140; demand('h9','c1')  = 200; demand('h10','c1') = 180; demand('h11','c1') = 180;
demand('h12','c1') = 240; demand('h13','c1') = 220; demand('h14','c1') = 220; demand('h15','c1') = 230;
demand('h16','c1') = 250; demand('h17','c1') = 260; demand('h18','c1') = 230; demand('h19','c1') = 250;
demand('h20','c1') = 260; demand('h21','c1') = 250; demand('h22','c1') = 220; demand('h23','c1') = 190;

* Company 2
demand('h0','c2')  = 70;  demand('h1','c2')  = 40;  demand('h2','c2')  = 20;  demand('h3','c2')  = 30;
demand('h4','c2')  = 50;  demand('h5','c2')  = 40;  demand('h6','c2')  = 40;  demand('h7','c2')  = 90;
demand('h8','c2')  = 110; demand('h9','c2')  = 110; demand('h10','c2') = 110; demand('h11','c2') = 130;
demand('h12','c2') = 150; demand('h13','c2') = 160; demand('h14','c2') = 120; demand('h15','c2') = 130;
demand('h16','c2') = 160; demand('h17','c2') = 180; demand('h18','c2') = 200; demand('h19','c2') = 170;
demand('h20','c2') = 140; demand('h21','c2') = 110; demand('h22','c2') = 100; demand('h23','c2') = 90;

* Coverage mapping Q(i,k): 6-hour cyclic window (wraps around midnight)
Q('h0','h18')=yes; Q('h0','h19')=yes; Q('h0','h20')=yes; Q('h0','h22')=yes; Q('h0','h23')=yes; Q('h0','h0')=yes;
Q('h1','h19')=yes; Q('h1','h20')=yes; Q('h1','h21')=yes; Q('h1','h23')=yes; Q('h1','h0')=yes;  Q('h1','h1')=yes;
Q('h2','h20')=yes; Q('h2','h21')=yes; Q('h2','h22')=yes; Q('h2','h0')=yes; Q('h2','h1')=yes;  Q('h2','h2')=yes;
Q('h3','h21')=yes; Q('h3','h22')=yes; Q('h3','h23')=yes; Q('h3','h1')=yes; Q('h3','h2')=yes;  Q('h3','h3')=yes;
Q('h4','h22')=yes; Q('h4','h23')=yes; Q('h4','h0')=yes;  Q('h4','h2')=yes; Q('h4','h3')=yes;  Q('h4','h4')=yes;
Q('h5','h23')=yes; Q('h5','h0')=yes;  Q('h5','h1')=yes;  Q('h5','h3')=yes; Q('h5','h4')=yes;  Q('h5','h5')=yes;
Q('h6','h0')=yes;  Q('h6','h1')=yes;  Q('h6','h2')=yes;  Q('h6','h4')=yes; Q('h6','h5')=yes;  Q('h6','h6')=yes;
Q('h7','h1')=yes;  Q('h7','h2')=yes;  Q('h7','h3')=yes;  Q('h7','h5')=yes; Q('h7','h6')=yes;  Q('h7','h7')=yes;
Q('h8','h2')=yes;  Q('h8','h3')=yes;  Q('h8','h4')=yes;  Q('h8','h6')=yes; Q('h8','h7')=yes;  Q('h8','h8')=yes;
Q('h9','h3')=yes;  Q('h9','h4')=yes;  Q('h9','h5')=yes;  Q('h9','h7')=yes; Q('h9','h8')=yes;  Q('h9','h9')=yes;
Q('h10','h4')=yes; Q('h10','h5')=yes; Q('h10','h6')=yes; Q('h10','h8')=yes;
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Q('h10','h9')=yes; Q('h10','h10')=yes;
Q('h11','h5')=yes; Q('h11','h6')=yes; Q('h11','h7')=yes; Q('h11','h9')=yes;
Q('h11','h10')=yes;Q('h11','h11')=yes;
Q('h12','h6')=yes; Q('h12','h7')=yes; Q('h12','h8')=yes;
Q('h12','h10')=yes;Q('h12','h11')=yes;Q('h12','h12')=yes;
Q('h13','h7')=yes; Q('h13','h8')=yes; Q('h13','h9')=yes;
Q('h13','h11')=yes;Q('h13','h12')=yes;Q('h13','h13')=yes;
Q('h14','h8')=yes; Q('h14','h9')=yes;
Q('h14','h10')=yes;Q('h14','h12')=yes;Q('h14','h13')=yes;Q('h14','h14')=yes;
Q('h15','h9')=yes;
Q('h15','h10')=yes;Q('h15','h11')=yes;Q('h15','h13')=yes;Q('h15','h14')=yes;Q('h15',
'h15')=yes;
Q('h16','h10')=yes;Q('h16','h11')=yes;Q('h16','h12')=yes;Q('h16','h14')=yes;Q('h16',
'h15')=yes;Q('h16','h16')=yes;
Q('h17','h11')=yes;Q('h17','h12')=yes;Q('h17','h13')=yes;Q('h17','h15')=yes;Q('h17',
'h16')=yes;Q('h17','h17')=yes;
Q('h18','h12')=yes;Q('h18','h13')=yes;Q('h18','h14')=yes;Q('h18','h16')=yes;Q('h18',
'h17')=yes;Q('h18','h18')=yes;
Q('h19','h13')=yes;Q('h19','h14')=yes;Q('h19','h15')=yes;Q('h19','h17')=yes;Q('h19',
'h18')=yes;Q('h19','h19')=yes;
Q('h20','h14')=yes;Q('h20','h15')=yes;Q('h20','h16')=yes;Q('h20','h18')=yes;Q('h20',
'h19')=yes;Q('h20','h20')=yes;
Q('h21','h15')=yes;Q('h21','h16')=yes;Q('h21','h17')=yes;Q('h21','h19')=yes;Q('h21',
'h20')=yes;Q('h21','h21')=yes;
Q('h22','h16')=yes;Q('h22','h17')=yes;Q('h22','h18')=yes;Q('h22','h20')=yes;Q('h22',
'h21')=yes;Q('h22','h22')=yes;
Q('h23','h17')=yes;Q('h23','h18')=yes;Q('h23','h19')=yes;Q('h23','h21')=yes;Q('h23',
'h22')=yes;Q('h23','h23')=yes;

* ==========================================================
* PART 2: LP relaxation + ranging (solver sensitivity)
*   - sensitivity/ranging works only for LP (continuous variables)
* ==========================================================

Positive Variables
    xLP(h,j)
    LLP(h,j);

Variable zLP;
```

```
  99  Equations
 100      objLP
 101      cap_per_hourLP(h)
 102      cap_totalLP(h)
 103      shortageLP(h,j)
 104      shortage_capLP(h,j)
 105      shortage_totalLP(j)
 106      min_coverageLP(h);
 107
 108  objLP..
 109      zLP =e=
 110          sum((h,j), cost(h)*xLP(h,j))
 111        + 60*sum(i$peak(i), LLP(i,'c1'))
 112        + 30*sum(i$off(i),  LLP(i,'c1'))
 113        + 45*sum(i,         LLP(i,'c2'));
 114
 115  cap_per_hourLP(h)..
 116      sum(j, xLP(h,j)) =l= capHour(h);
 117
 118  cap_totalLP(h)..
 119      sum((k,j)$Q(h,k), xLP(k,j)) =l= capTotal(h);
 120
 121  shortageLP(h,j)..
 122      demand(h,j) - sum(k$Q(h,k), xLP(k,j)) =l= LLP(h,j);
 123
 124  shortage_capLP(h,j)..
 125      LLP(h,j) =l= maxShort(h,j);
 126
 127  shortage_totalLP(j)..
 128      sum(h, LLP(h,j)) =l= totShort(j);
 129
 130  min_coverageLP(h)..
 131      sum((k,j)$Q(h,k), xLP(k,j)) =g= minCov(h);
 132
 133  Model Telecom_LP
/objLP,cap_per_hourLP,cap_totalLP,shortageLP,shortage_capLP,shortage_totalLP,min_cov
erageLP/;
 134
 139  Telecom_LP.optfile = 1;
 140
 141  Solve Telecom_LP using LP minimizing zLP;
 142
 143  display "PART 2 (LP relaxation) objective", zLP.l;


COMPILATION TIME     =        0.000 SECONDS      3 MB  52.5.0 9bc771bf WEX-WEI
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Equation Listing   SOLVE Telecom_LP Using LP From line 141

```
---- objLP  =E=

objLP..  - 189*xLP(h0,c1) - 189*xLP(h0,c2) - 189*xLP(h1,c1) - 189*xLP(h1,c2) -
189*xLP(h2,c1) - 189*xLP(h2,c2) - 180*xLP(h3,c1) - 180*xLP(h3,c2) - 171*xLP(h4,c1) -
171*xLP(h4,c2) - 162*xLP(h5,c1) - 162*xLP(h5,c2) - 153*xLP(h6,c1) - 153*xLP(h6,c2) -
144*xLP(h7,c1) - 144*xLP(h7,c2) - 135*xLP(h8,c1) - 135*xLP(h8,c2) - 126*xLP(h9,c1) -
126*xLP(h9,c2) - 126*xLP(h10,c1) - 126*xLP(h10,c2) - 135*xLP(h11,c1) -
135*xLP(h11,c2) - 144*xLP(h12,c1) - 144*xLP(h12,c2) - 153*xLP(h13,c1) -
153*xLP(h13,c2) - 162*xLP(h14,c1) - 162*xLP(h14,c2) - 171*xLP(h15,c1) -
171*xLP(h15,c2) - 180*xLP(h16,c1) - 180*xLP(h16,c2) - 189*xLP(h17,c1) -
189*xLP(h17,c2) - 189*xLP(h18,c1) - 189*xLP(h18,c2) - 189*xLP(h19,c1) -
189*xLP(h19,c2) - 189*xLP(h20,c1) - 189*xLP(h20,c2) - 189*xLP(h21,c1) -
189*xLP(h21,c2) - 189*xLP(h22,c1) - 189*xLP(h22,c2) - 189*xLP(h23,c1) -
189*xLP(h23,c2) - 30*LLP(h0,c1) - 45*LLP(h0,c2) - 30*LLP(h1,c1) - 45*LLP(h1,c2) -
30*LLP(h2,c1) - 45*LLP(h2,c2) - 30*LLP(h3,c1) - 45*LLP(h3,c2) - 30*LLP(h4,c1) -
45*LLP(h4,c2) - 30*LLP(h5,c1) - 45*LLP(h5,c2) - 30*LLP(h6,c1) - 45*LLP(h6,c2) -
60*LLP(h7,c1) - 45*LLP(h7,c2) - 60*LLP(h8,c1) - 45*LLP(h8,c2) - 60*LLP(h9,c1) -
45*LLP(h9,c2) - 60*LLP(h10,c1) - 45*LLP(h10,c2) - 60*LLP(h11,c1) - 45*LLP(h11,c2) -
60*LLP(h12,c1) - 45*LLP(h12,c2) - 60*LLP(h13,c1) - 45*LLP(h13,c2) - 60*LLP(h14,c1) -
45*LLP(h14,c2) - 60*LLP(h15,c1) - 45*LLP(h15,c2) - 60*LLP(h16,c1) - 45*LLP(h16,c2) -
60*LLP(h17,c1) - 45*LLP(h17,c2) - 30*LLP(h18,c1) - 45*LLP(h18,c2) - 30*LLP(h19,c1) -
45*LLP(h19,c2) - 30*LLP(h20,c1) - 45*LLP(h20,c2) - 30*LLP(h21,c1) - 45*LLP(h21,c2) -
30*LLP(h22,c1) - 45*LLP(h22,c2) - 30*LLP(h23,c1) - 45*LLP(h23,c2) + zLP =E= 0 ; (LHS
= 0)


---- cap_per_hourLP  =L=

cap_per_hourLP(h0)..  xLP(h0,c1) + xLP(h0,c2) =L= 100 ; (LHS = 0)

cap_per_hourLP(h1)..  xLP(h1,c1) + xLP(h1,c2) =L= 100 ; (LHS = 0)

cap_per_hourLP(h2)..  xLP(h2,c1) + xLP(h2,c2) =L= 100 ; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED


---- cap_totalLP  =L=

cap_totalLP(h0)..  xLP(h0,c1) + xLP(h0,c2) + xLP(h18,c1) + xLP(h18,c2) + xLP(h19,c1)
+ xLP(h19,c2) + xLP(h20,c1) + xLP(h20,c2) + xLP(h22,c1) + xLP(h22,c2) + xLP(h23,c1)
+ xLP(h23,c2) =L= 420 ; (LHS = 0)

cap_totalLP(h1)..  xLP(h0,c1) + xLP(h0,c2) + xLP(h1,c1) + xLP(h1,c2) + xLP(h19,c1) +
xLP(h19,c2) + xLP(h20,c1) + xLP(h20,c2) + xLP(h21,c1) + xLP(h21,c2) + xLP(h23,c1) +
xLP(h23,c2) =L= 420 ; (LHS = 0)

cap_totalLP(h2)..  xLP(h0,c1) + xLP(h0,c2) + xLP(h1,c1) + xLP(h1,c2) + xLP(h2,c1) +
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xLP(h2,c2) + xLP(h20,c1) + xLP(h20,c2) + xLP(h21,c1) + xLP(h21,c2) + xLP(h22,c1) +
xLP(h22,c2) =L= 420 ; (LHS = 0)

REMAINING 21 ENTRIES SKIPPED


---- shortageLP  =L=

shortageLP(h0,c1)..  - xLP(h0,c1) - xLP(h18,c1) - xLP(h19,c1) - xLP(h20,c1) -
xLP(h22,c1) - xLP(h23,c1) - LLP(h0,c1) =L= -120 ; (LHS = 0, INFES = 120 ****)

shortageLP(h0,c2)..  - xLP(h0,c2) - xLP(h18,c2) - xLP(h19,c2) - xLP(h20,c2) -
xLP(h22,c2) - xLP(h23,c2) - LLP(h0,c2) =L= -70 ; (LHS = 0, INFES = 70 ****)

shortageLP(h1,c1)..  - xLP(h0,c1) - xLP(h1,c1) - xLP(h19,c1) - xLP(h20,c1) -
xLP(h21,c1) - xLP(h23,c1) - LLP(h1,c1) =L= -95 ; (LHS = 0, INFES = 95 ****)

REMAINING 45 ENTRIES SKIPPED


---- shortage_capLP  =L=

shortage_capLP(h0,c1)..  LLP(h0,c1) =L= 10 ; (LHS = 0)

shortage_capLP(h0,c2)..  LLP(h0,c2) =L= 10 ; (LHS = 0)

shortage_capLP(h1,c1)..  LLP(h1,c1) =L= 10 ; (LHS = 0)

REMAINING 45 ENTRIES SKIPPED


---- shortage_totalLP  =L=

shortage_totalLP(c1)..  LLP(h0,c1) + LLP(h1,c1) + LLP(h2,c1) + LLP(h3,c1) +
LLP(h4,c1) + LLP(h5,c1) + LLP(h6,c1) + LLP(h7,c1) + LLP(h8,c1) + LLP(h9,c1) +
LLP(h10,c1) + LLP(h11,c1) + LLP(h12,c1) + LLP(h13,c1) + LLP(h14,c1) + LLP(h15,c1) +
LLP(h16,c1) + LLP(h17,c1) + LLP(h18,c1) + LLP(h19,c1) + LLP(h20,c1) + LLP(h21,c1) +
LLP(h22,c1) + LLP(h23,c1) =L= 40 ; (LHS = 0)

shortage_totalLP(c2)..  LLP(h0,c2) + LLP(h1,c2) + LLP(h2,c2) + LLP(h3,c2) +
LLP(h4,c2) + LLP(h5,c2) + LLP(h6,c2) + LLP(h7,c2) + LLP(h8,c2) + LLP(h9,c2) +
LLP(h10,c2) + LLP(h11,c2) + LLP(h12,c2) + LLP(h13,c2) + LLP(h14,c2) + LLP(h15,c2) +
LLP(h16,c2) + LLP(h17,c2) + LLP(h18,c2) + LLP(h19,c2) + LLP(h20,c2) + LLP(h21,c2) +
LLP(h22,c2) + LLP(h23,c2) =L= 30 ; (LHS = 0)


---- min_coverageLP  =G=

min_coverageLP(h0)..  xLP(h0,c1) + xLP(h0,c2) + xLP(h18,c1) + xLP(h18,c2) +
xLP(h19,c1) + xLP(h19,c2) + xLP(h20,c1) + xLP(h20,c2) + xLP(h22,c1) + xLP(h22,c2) +
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```
xLP(h23,c1) + xLP(h23,c2) =G= 80 ; (LHS = 0, INFES = 80 ****)

min_coverageLP(h1)..  xLP(h0,c1) + xLP(h0,c2) + xLP(h1,c1) + xLP(h1,c2) +
xLP(h19,c1) + xLP(h19,c2) + xLP(h20,c1) + xLP(h20,c2) + xLP(h21,c1) + xLP(h21,c2) +
xLP(h23,c1) + xLP(h23,c2) =G= 80 ; (LHS = 0, INFES = 80 ****)

min_coverageLP(h2)..  xLP(h0,c1) + xLP(h0,c2) + xLP(h1,c1) + xLP(h1,c2) + xLP(h2,c1)
+ xLP(h2,c2) + xLP(h20,c1) + xLP(h20,c2) + xLP(h21,c1) + xLP(h21,c2) + xLP(h22,c1) +
xLP(h22,c2) =G= 80 ; (LHS = 0, INFES = 80 ****)

REMAINING 21 ENTRIES SKIPPED
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Column Listing      SOLVE Telecom_LP Using LP From line 141


---- xLP

xLP(h0,c1)
                (.LO, .L, .UP, .M = 0, 0, +INF, 0)
     -189       objLP
        1       cap_per_hourLP(h0)
        1       cap_totalLP(h0)
        1       cap_totalLP(h1)
        1       cap_totalLP(h2)
        1       cap_totalLP(h4)
        1       cap_totalLP(h5)
        1       cap_totalLP(h6)
       -1       shortageLP(h0,c1)
       -1       shortageLP(h1,c1)
       -1       shortageLP(h2,c1)
       -1       shortageLP(h4,c1)
       -1       shortageLP(h5,c1)
       -1       shortageLP(h6,c1)
        1       min_coverageLP(h0)
        1       min_coverageLP(h1)
        1       min_coverageLP(h2)
        1       min_coverageLP(h4)
        1       min_coverageLP(h5)
        1       min_coverageLP(h6)

xLP(h0,c2)
                (.LO, .L, .UP, .M = 0, 0, +INF, 0)
     -189       objLP
        1       cap_per_hourLP(h0)
        1       cap_totalLP(h0)
        1       cap_totalLP(h1)
        1       cap_totalLP(h2)
```

```
         1       cap_totalLP(h4)
         1       cap_totalLP(h5)
         1       cap_totalLP(h6)
        -1       shortageLP(h0,c2)
        -1       shortageLP(h1,c2)
        -1       shortageLP(h2,c2)
        -1       shortageLP(h4,c2)
        -1       shortageLP(h5,c2)
        -1       shortageLP(h6,c2)
         1       min_coverageLP(h0)
         1       min_coverageLP(h1)
         1       min_coverageLP(h2)
         1       min_coverageLP(h4)
         1       min_coverageLP(h5)
         1       min_coverageLP(h6)

xLP(h1,c1)
                 (.LO, .L, .UP, .M = 0, 0, +INF, 0)
      -189       objLP
         1       cap_per_hourLP(h1)
         1       cap_totalLP(h1)
         1       cap_totalLP(h2)
         1       cap_totalLP(h3)
         1       cap_totalLP(h5)
         1       cap_totalLP(h6)
         1       cap_totalLP(h7)
        -1       shortageLP(h1,c1)
        -1       shortageLP(h2,c1)
        -1       shortageLP(h3,c1)
        -1       shortageLP(h5,c1)
        -1       shortageLP(h6,c1)
        -1       shortageLP(h7,c1)
         1       min_coverageLP(h1)
         1       min_coverageLP(h2)
         1       min_coverageLP(h3)
         1       min_coverageLP(h5)
         1       min_coverageLP(h6)
         1       min_coverageLP(h7)

REMAINING 45 ENTRIES SKIPPED

---- LLP

LLP(h0,c1)
                 (.LO, .L, .UP, .M = 0, 0, +INF, 0)
       -30       objLP
        -1       shortageLP(h0,c1)
         1       shortage_capLP(h0,c1)
         1       shortage_totalLP(c1)
```

```
LLP(h0,c2)
                (.LO, .L, .UP, .M = 0, 0, +INF, 0)
      -45       objLP
       -1       shortageLP(h0,c2)
        1       shortage_capLP(h0,c2)
        1       shortage_totalLP(c2)

LLP(h1,c1)
                (.LO, .L, .UP, .M = 0, 0, +INF, 0)
      -30       objLP
       -1       shortageLP(h1,c1)
        1       shortage_capLP(h1,c1)
        1       shortage_totalLP(c1)

REMAINING 45 ENTRIES SKIPPED

---- zLP

zLP
                (.LO, .L, .UP, .M = -INF, 0, +INF, 0)
        1       objLP
```


Range Statistics   SOLVE Telecom_LP Using LP From line 141

```
RANGE STATISTICS (ABSOLUTE NON-ZERO FINITE VALUES)

RHS       [min, max] : [ 1.000E+01, 4.200E+02] - Zero values observed as well
Bound     [min, max] : [        NA,        NA] - Zero values observed as well
Matrix    [min, max] : [ 1.000E+00, 1.890E+02]
```


Model Statistics   SOLVE Telecom_LP Using LP From line 141

```
MODEL STATISTICS

BLOCKS OF EQUATIONS           7     SINGLE EQUATIONS          171
BLOCKS OF VARIABLES           3     SINGLE VARIABLES           97
NON ZERO ELEMENTS         1,153


GENERATION TIME      =        0.031 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI
```

```
G e n e r a l   A l g e b r a i c   M o d e l i n g   S y s t e m
Solution Report     SOLVE Telecom_LP Using LP From line 141


               S O L V E      S U M M A R Y

     MODEL   Telecom_LP          OBJECTIVE  zLP
     TYPE    LP                  DIRECTION  MINIMIZE
     SOLVER  CPLEX               FROM LINE  141

**** SOLVER STATUS     1 Normal Completion
**** MODEL STATUS      1 Optimal
**** OBJECTIVE VALUE           184811.6667

 RESOURCE USAGE, LIMIT          0.016 10000000000.000
 ITERATION COUNT, LIMIT        79    2147483647
*** This solver runs with a demo license. No commercial use.

Reading parameter(s) from "C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\cplex.opt"
>>  objrng all
>>  rhsrng all
Finished reading from "C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\cplex.opt"

--- GMO setup time: 0.00s
--- GMO memory 0.53 Mb (peak 0.53 Mb)
--- Dictionary memory 0.00 Mb
--- Cplex 22.1.2.0 link memory 0.00 Mb (peak 0.02 Mb)
--- Starting Cplex


--- LP status (1): optimal.
--- Cplex Time: 0.02sec (det. 1.07 ticks)

--- Start ranging / sensitivity analysis...
--- Right-hand-side ranging...

EQUATION NAME                           LOWER           CURRENT             UPPER
-------------                           -----           -------             -----
objLP                                    -INF                NA              +INF
=C
cap_per_hourLP(h0)                    8.20988               100              +INF
=C
cap_per_hourLP(h1)                    4.38272               100              +INF
=C
cap_per_hourLP(h2)                    30.6173               100              +INF
=C
cap_per_hourLP(h3)                    12.5926               100              +INF
=C
cap_per_hourLP(h4)                    22.0988               100              +INF
=C
```

cap_per_hourLP(h5) =C	32.716	100	+INF
cap_per_hourLP(h6) =C	19.6914	100	+INF
cap_per_hourLP(h7) =C	97.6389	100	101.389
cap_per_hourLP(h8) =C	65	100	+INF
cap_per_hourLP(h9) =C	77.5926	100	+INF
cap_per_hourLP(h10) =C	72.9012	100	+INF
cap_per_hourLP(h11) =C	43.7037	100	+INF
cap_per_hourLP(h12) =C	88.7037	100	+INF
cap_per_hourLP(h13) =C	37.716	100	+INF
cap_per_hourLP(h14) =C	43.7037	100	+INF
cap_per_hourLP(h15) =C	97.2222	100	104.722
cap_per_hourLP(h16) =C	65.4938	100	+INF
cap_per_hourLP(h17) =C	84.3827	100	+INF
cap_per_hourLP(h18) =C	82.716	100	104.074
cap_per_hourLP(h19) =C	49.8148	100	+INF
cap_per_hourLP(h20) =C	30.9877	100	+INF
cap_per_hourLP(h21) =C	29.321	100	+INF
cap_per_hourLP(h22) =C	6.48148	100	+INF
cap_per_hourLP(h23) =C	0	100	+INF
cap_totalLP(h0) =C	195.494	420	+INF
cap_totalLP(h1) =C	122.716	420	+INF
cap_totalLP(h2) =C	110	420	+INF
cap_totalLP(h3) =C	83.3951	420	+INF
cap_totalLP(h4) =C	80	420	+INF
cap_totalLP(h5) =C	80	420	+INF

cap_totalLP(h6) =C	117.716	420	+INF
cap_totalLP(h7) =C	200	420	+INF
cap_totalLP(h8) =C	250	420	+INF
cap_totalLP(h9) =C	310	420	+INF
cap_totalLP(h10) =C	290	420	+INF
cap_totalLP(h11) =C	346.605	420	+INF
cap_totalLP(h12) =C	390	420	+INF
cap_totalLP(h13) =C	412.716	420	+INF
cap_totalLP(h14) =C	385.617	420	+INF
cap_totalLP(h15) =C	375.617	420	+INF
cap_totalLP(h16) =C	414.506	420	+INF
cap_totalLP(h17) =C	420	420	420.298
cap_totalLP(h18) =C	418.333	420	420.379
cap_totalLP(h19) =C	415.617	420	+INF
cap_totalLP(h20) =C	390	420	+INF
cap_totalLP(h21) =C	360	420	+INF
cap_totalLP(h22) =C	316.667	420	+INF
cap_totalLP(h23) =C	270	420	+INF
shortageLP(h0.c1) =C	-125.494	-120	+INF
shortageLP(h0.c2) =C	-70	-70	+INF
shortageLP(h1.c1) =C	-98.2883	-95	-94.359
shortageLP(h1.c2) =C	-41.3077	-40	-38.6842
shortageLP(h2.c1) =C	-90.5556	-90	-86.6667
shortageLP(h2.c2) =C	-20.5319	-20	-17.5
shortageLP(h3.c1) =C	-53.3951	-50	+INF

shortageLP(h3.c2) =C	-30.6579	-30	-28.1544
shortageLP(h4.c1) =C	-30	-30	+INF
shortageLP(h4.c2) =C	-50	-50	-50
shortageLP(h5.c1) =C	-38.9506	-30	+INF
shortageLP(h5.c2) =C	-41.0494	-40	+INF
shortageLP(h6.c1) =C	-60.7576	-60	-56.6667
shortageLP(h6.c2) =C	-57.716	-40	+INF
shortageLP(h7.c1) =C	-120.667	-110	-109.359
shortageLP(h7.c2) =C	-91.337	-90	-89.359
shortageLP(h8.c1) =C	-146.852	-140	-139.074
shortageLP(h8.c2) =C	-110	-110	-109.444
shortageLP(h9.c1) =C	-201.042	-200	-194.394
shortageLP(h9.c2) =C	-110.338	-110	-107.875
shortageLP(h10.c1) =C	-182.778	-180	-176.667
shortageLP(h10.c2) =C	-113.174	-110	-108.365
shortageLP(h11.c1) =C	-216.605	-180	+INF
shortageLP(h11.c2) =C	-131.25	-130	-128.119
shortageLP(h12.c1) =C	-241.042	-240	-235.805
shortageLP(h12.c2) =C	-150.385	-150	-150
shortageLP(h13.c1) =C	-252.716	-220	+INF
shortageLP(h13.c2) =C	-161.667	-160	-159.731
shortageLP(h14.c1) =C	-223.333	-220	-219.167
shortageLP(h14.c2) =C	-165.617	-120	+INF
shortageLP(h15.c1) =C	-237.417	-230	-223.917
shortageLP(h15.c2) =C	-145.617	-130	+INF

shortageLP(h16.c1) =C	-254.506	-250	+INF
shortageLP(h16.c2) =C	-160	-160	-159.702
shortageLP(h17.c1) =C	-260	-260	-259.762
shortageLP(h17.c2) =C	-180	-180	-178.684
shortageLP(h18.c1) =C	-233.288	-230	-229.359
shortageLP(h18.c2) =C	-204.62	-200	-198.684
shortageLP(h19.c1) =C	-253.395	-250	-244.506
shortageLP(h19.c2) =C	-174.62	-170	-168.684
shortageLP(h20.c1) =C	-264.167	-260	-252.917
shortageLP(h20.c2) =C	-140.446	-140	-140
shortageLP(h21.c1) =C	-250.926	-250	-246.62
shortageLP(h21.c2) =C	-114.722	-110	-108.651
shortageLP(h22.c1) =C	-223.288	-220	-219.359
shortageLP(h22.c2) =C	-101.667	-100	-99.6667
shortageLP(h23.c1) =C	-194.167	-190	-187.135
shortageLP(h23.c2) =C	-92.9365	-90	-87.2222
shortage_capLP(h0.c1) =C	0	10	+INF
shortage_capLP(h0.c2) =C	0	10	+INF
shortage_capLP(h1.c1) =C	3.33333	10	+INF
shortage_capLP(h1.c2) =C	8.95062	10	+INF
shortage_capLP(h2.c1) =C	0	10	+INF
shortage_capLP(h2.c2) =C	0	10	+INF
shortage_capLP(h3.c1) =C	0	10	+INF
shortage_capLP(h3.c2) =C	0	10	+INF
shortage_capLP(h4.c1) =C	0	10	+INF

shortage_capLP(h4.c2) =C	0	10	+INF
shortage_capLP(h5.c1) =C	0	10	+INF
shortage_capLP(h5.c2) =C	0	10	+INF
shortage_capLP(h6.c1) =C	0	10	+INF
shortage_capLP(h6.c2) =C	0	10	+INF
shortage_capLP(h7.c1) =C	0	10	+INF
shortage_capLP(h7.c2) =C	0	10	+INF
shortage_capLP(h8.c1) =C	0	10	+INF
shortage_capLP(h8.c2) =C	0	10	+INF
shortage_capLP(h9.c1) =C	0	10	+INF
shortage_capLP(h9.c2) =C	0	10	+INF
shortage_capLP(h10.c1) =C	0	10	+INF
shortage_capLP(h10.c2) =C	0	10	+INF
shortage_capLP(h11.c1) =C	0	10	+INF
shortage_capLP(h11.c2) =C	0	10	+INF
shortage_capLP(h12.c1) =C	0	10	+INF
shortage_capLP(h12.c2) =C	0	10	+INF
shortage_capLP(h13.c1) =C	0	10	+INF
shortage_capLP(h13.c2) =C	0	10	+INF
shortage_capLP(h14.c1) =C	0	10	+INF
shortage_capLP(h14.c2) =C	0	10	+INF
shortage_capLP(h15.c1) =C	0	10	+INF
shortage_capLP(h15.c2) =C	0	10	+INF
shortage_capLP(h16.c1) =C	0	10	+INF
shortage_capLP(h16.c2) =C	0	10	+INF

```
shortage_capLP(h17.c1)                        10              10          10.3788
=C
shortage_capLP(h17.c2)                        10              10             +INF
=C
shortage_capLP(h18.c1)                   3.33333              10             +INF
=C
shortage_capLP(h18.c2)                   6.66667              10             +INF
=C
shortage_capLP(h19.c1)                         0              10             +INF
=C
shortage_capLP(h19.c2)                   4.38272              10             +INF
=C
shortage_capLP(h20.c1)                   9.44444              10          12.3856
=C
shortage_capLP(h20.c2)                         0              10             +INF
=C
shortage_capLP(h21.c1)                         0              10             +INF
=C
shortage_capLP(h21.c2)                         0              10             +INF
=C
shortage_capLP(h22.c1)                   3.33333              10             +INF
=C
shortage_capLP(h22.c2)                         0              10             +INF
=C
shortage_capLP(h23.c1)                   9.44444              10          11.7974
=C
shortage_capLP(h23.c2)                         0              10             +INF
=C
shortage_totalLP(c1)                     36.7117              40           40.641
=C
shortage_totalLP(c2)                     25.3797              30          31.3158
=C
min_coverageLP(h0)                          -INF              80          195.494
=C
min_coverageLP(h1)                          -INF              80          122.716
=C
min_coverageLP(h2)                          -INF              80              110
=C
min_coverageLP(h3)                          -INF              80          83.3951
=C
min_coverageLP(h4)                            80              80          83.5417
=C
min_coverageLP(h5)                       79.0741              80          86.8519
=C
min_coverageLP(h6)                          -INF              80          117.716
=C
min_coverageLP(h7)                          -INF              80              200
=C
min_coverageLP(h8)                          -INF              80              250
=C
```

```
min_coverageLP(h9)                     -INF          80         310
=C
min_coverageLP(h10)                    -INF          80         290
=C
min_coverageLP(h11)                    -INF          80     346.605
=C
min_coverageLP(h12)                    -INF          80         390
=C
min_coverageLP(h13)                    -INF          80     412.716
=C
min_coverageLP(h14)                    -INF          80     385.617
=C
min_coverageLP(h15)                    -INF          80     375.617
=C
min_coverageLP(h16)                    -INF          80     414.506
=C
min_coverageLP(h17)                    -INF          80         420
=C
min_coverageLP(h18)                    -INF          80         420
=C
min_coverageLP(h19)                    -INF          80     415.617
=C
min_coverageLP(h20)                    -INF          80         390
=C
min_coverageLP(h21)                    -INF          80         360
=C
min_coverageLP(h22)                    -INF          80     316.667
=C
min_coverageLP(h23)                    -INF          80         270
=C

--- Objective ranging...

VARIABLE NAME                         LOWER     CURRENT       UPPER
-------------                         -----     -------       -----
xLP(h0.c1)                              189         189         189
=C
xLP(h0.c2)                              189         189         189
=C
xLP(h1.c1)                              189         189        +INF
=C
xLP(h1.c2)                              189         189         189
=C
xLP(h2.c1)                              189         189         189
=C
xLP(h2.c2)                              189         189         189
=C
xLP(h3.c1)                              180         180         180
=C
xLP(h3.c2)                              180         180         180
```

=C xLP(h4.c1)	171	171	+INF
=C xLP(h4.c2)	171	171	171
=C xLP(h5.c1)	162	162	162
=C xLP(h5.c2)	162	162	+INF
=C xLP(h6.c1)	153	153	153
=C xLP(h6.c2)	153	153	153
=C xLP(h7.c1)	144	144	144
=C xLP(h7.c2)	144	144	144
=C xLP(h8.c1)	135	135	135
=C xLP(h8.c2)	135	135	135
=C xLP(h9.c1)	126	126	126
=C xLP(h9.c2)	126	126	126
=C xLP(h10.c1)	126	126	126
=C xLP(h10.c2)	126	126	126
=C xLP(h11.c1)	130.5	135	135
=C xLP(h11.c2)	135	135	+INF
=C xLP(h12.c1)	144	144	144
=C xLP(h12.c2)	144	144	144
=C xLP(h13.c1)	153	153	+INF
=C xLP(h13.c2)	153	153	153
=C xLP(h14.c1)	162	162	162
=C xLP(h14.c2)	162	162	162
=C xLP(h15.c1)	171	171	171
=C xLP(h15.c2)	171	171	171
=C xLP(h16.c1)	180	180	180

=C xLP(h16.c2)	180	180	180
=C xLP(h17.c1)	189	189	189
=C xLP(h17.c2)	189	189	189
=C xLP(h18.c1)	189	189	189
=C xLP(h18.c2)	172	189	189
=C xLP(h19.c1)	189	189	189
=C xLP(h19.c2)	189	189	189
=C xLP(h20.c1)	189	189	189
=C xLP(h20.c2)	189	189	+INF
=C xLP(h21.c1)	189	189	189
=C xLP(h21.c2)	189	189	+INF
=C xLP(h22.c1)	189	189	189
=C xLP(h22.c2)	189	189	+INF
=C xLP(h23.c1)	183	189	+INF
=C xLP(h23.c2)	183	189	+INF
=C LLP(h0.c1)	-17	30	+INF
=C LLP(h0.c2)	-16.6667	45	+INF
=C LLP(h1.c1)	27	30	38
=C LLP(h1.c2)	45	45	45
=C LLP(h2.c1)	7	30	+INF
=C LLP(h2.c2)	25	45	+INF
=C LLP(h3.c1)	-17	30	+INF
=C LLP(h3.c2)	-5	45	+INF
=C LLP(h4.c1)	-17	30	+INF
=C LLP(h4.c2)	-5	45	+INF

=C			
LLP(h5.c1)	-17	30	+INF
=C			
LLP(h5.c2)	-16.6667	45	+INF
=C			
LLP(h6.c1)	27	30	+INF
=C			
LLP(h6.c2)	-16.6667	45	+INF
=C			
LLP(h7.c1)	39	60	+INF
=C			
LLP(h7.c2)	39.3333	45	+INF
=C			
LLP(h8.c1)	-8	60	+INF
=C			
LLP(h8.c2)	-4.66667	45	+INF
=C			
LLP(h9.c1)	24	60	+INF
=C			
LLP(h9.c2)	-2	45	+INF
=C			
LLP(h10.c1)	-14	60	+INF
=C			
LLP(h10.c2)	42	45	+INF
=C			
LLP(h11.c1)	-17	60	+INF
=C			
LLP(h11.c2)	-2	45	+INF
=C			
LLP(h12.c1)	24	60	+INF
=C			
LLP(h12.c2)	-5	45	+INF
=C			
LLP(h13.c1)	-17	60	+INF
=C			
LLP(h13.c2)	21.3333	45	+INF
=C			
LLP(h14.c1)	24	60	+INF
=C			
LLP(h14.c2)	-16.6667	45	+INF
=C			
LLP(h15.c1)	24	60	+INF
=C			
LLP(h15.c2)	-16.6667	45	+INF
=C			
LLP(h16.c1)	-17	60	+INF
=C			
LLP(h16.c2)	24.3333	45	+INF
=C			
LLP(h17.c1)	-INF	60	68

```
=C
LLP(h17.c2)                    37              45             +INF
=C
LLP(h18.c1)                    22              30             38.5
=C
LLP(h18.c2)                  36.5              45               53
=C
LLP(h19.c1)                    -8              30             +INF
=C
LLP(h19.c2)                    45              45               45
=C
LLP(h20.c1)                  -INF              30               45
=C
LLP(h20.c2)                    19              45             +INF
=C
LLP(h21.c1)                    -8              30             +INF
=C
LLP(h21.c2)              -10.6667              45             +INF
=C
LLP(h22.c1)                    22              30               33
=C
LLP(h22.c2)               27.3333              45             +INF
=C
LLP(h23.c1)                  -INF              30               45
=C
LLP(h23.c2)               7.33333              45             +INF
=C
zLP                          -INF              NA             +INF
=C


Optimal solution found
Objective:       184811.666667


                           LOWER          LEVEL          UPPER         MARGINAL

---- EQU objLP               .              .              .             1.0000


---- EQU cap_per_hourLP

           LOWER          LEVEL          UPPER         MARGINAL

h0         -INF            8.2099       100.0000          .
h1         -INF            4.3827       100.0000          .
h2         -INF           30.6173       100.0000          .
h3         -INF           12.5926       100.0000          .
h4         -INF           22.0988       100.0000          .
h5         -INF           32.7160       100.0000          .
```

```
h6          -INF          19.6914       100.0000          .
h7          -INF         100.0000       100.0000       -3.0000
h8          -INF          65.0000       100.0000          .
h9          -INF          77.5926       100.0000          .
h10         -INF          72.9012       100.0000          .
h11         -INF          43.7037       100.0000          .
h12         -INF          88.7037       100.0000          .
h13         -INF          37.7160       100.0000          .
h14         -INF          43.7037       100.0000          .
h15         -INF         100.0000       100.0000       -3.0000
h16         -INF          65.4938       100.0000          .
h17         -INF          84.3827       100.0000          .
h18         -INF         100.0000       100.0000         EPS
h19         -INF          49.8148       100.0000          .
h20         -INF          30.9877       100.0000          .
h21         -INF          29.3210       100.0000          .
h22         -INF           6.4815       100.0000          .
h23         -INF             .          100.0000          .

---- EQU cap_totalLP

            LOWER          LEVEL          UPPER         MARGINAL

h0          -INF         195.4938       420.0000          .
h1          -INF         122.7160       420.0000          .
h2          -INF         110.0000       420.0000          .
h3          -INF          83.3951       420.0000          .
h4          -INF          80.0000       420.0000          .
h5          -INF          80.0000       420.0000          .
h6          -INF         117.7160       420.0000          .
h7          -INF         200.0000       420.0000          .
h8          -INF         250.0000       420.0000          .
h9          -INF         310.0000       420.0000          .
h10         -INF         290.0000       420.0000          .
h11         -INF         346.6049       420.0000          .
h12         -INF         390.0000       420.0000          .
h13         -INF         412.7160       420.0000          .
h14         -INF         385.6173       420.0000          .
h15         -INF         375.6173       420.0000          .
h16         -INF         414.5062       420.0000          .
h17         -INF         420.0000       420.0000      -32.0000
h18         -INF         420.0000       420.0000      -38.0000
h19         -INF         415.6173       420.0000          .
h20         -INF         390.0000       420.0000          .
h21         -INF         360.0000       420.0000          .
h22         -INF         316.6667       420.0000          .
h23         -INF         270.0000       420.0000          .

---- EQU shortageLP
```

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	-INF	-125.4938	-120.0000	.
h0 .c2	-INF	-70.0000	-70.0000	.
h1 .c1	-INF	-95.0000	-95.0000	-47.0000
h1 .c2	-INF	-40.0000	-40.0000	-61.6667
h2 .c1	-INF	-90.0000	-90.0000	-24.0000
h2 .c2	-INF	-20.0000	-20.0000	-41.6667
h3 .c1	-INF	-53.3951	-50.0000	.
h3 .c2	-INF	-30.0000	-30.0000	-11.6667
h4 .c1	-INF	-30.0000	-30.0000	.
h4 .c2	-INF	-50.0000	-50.0000	-11.6667
h5 .c1	-INF	-38.9506	-30.0000	.
h5 .c2	-INF	-41.0494	-40.0000	.
h6 .c1	-INF	-60.0000	-60.0000	-44.0000
h6 .c2	-INF	-57.7160	-40.0000	.
h7 .c1	-INF	-110.0000	-110.0000	-56.0000
h7 .c2	-INF	-90.0000	-90.0000	-56.0000
h8 .c1	-INF	-140.0000	-140.0000	-9.0000
h8 .c2	-INF	-110.0000	-110.0000	-12.0000
h9 .c1	-INF	-200.0000	-200.0000	-41.0000
h9 .c2	-INF	-110.0000	-110.0000	-14.6667
h10.c1	-INF	-180.0000	-180.0000	-3.0000
h10.c2	-INF	-110.0000	-110.0000	-58.6667
h11.c1	-INF	-216.6049	-180.0000	.
h11.c2	-INF	-130.0000	-130.0000	-14.6667
h12.c1	-INF	-240.0000	-240.0000	-41.0000
h12.c2	-INF	-150.0000	-150.0000	-11.6667
h13.c1	-INF	-252.7160	-220.0000	.
h13.c2	-INF	-160.0000	-160.0000	-38.0000
h14.c1	-INF	-220.0000	-220.0000	-41.0000
h14.c2	-INF	-165.6173	-120.0000	.
h15.c1	-INF	-230.0000	-230.0000	-41.0000
h15.c2	-INF	-145.6173	-130.0000	.
h16.c1	-INF	-254.5062	-250.0000	.
h16.c2	-INF	-160.0000	-160.0000	-41.0000
h17.c1	-INF	-260.0000	-260.0000	-85.0000
h17.c2	-INF	-180.0000	-180.0000	-61.6667
h18.c1	-INF	-230.0000	-230.0000	-47.0000
h18.c2	-INF	-200.0000	-200.0000	-61.6667
h19.c1	-INF	-250.0000	-250.0000	-9.0000
h19.c2	-INF	-170.0000	-170.0000	-61.6667
h20.c1	-INF	-260.0000	-260.0000	-62.0000
h20.c2	-INF	-140.0000	-140.0000	-35.6667
h21.c1	-INF	-250.0000	-250.0000	-9.0000
h21.c2	-INF	-110.0000	-110.0000	-6.0000
h22.c1	-INF	-220.0000	-220.0000	-47.0000
h22.c2	-INF	-100.0000	-100.0000	-44.0000
h23.c1	-INF	-190.0000	-190.0000	-62.0000
h23.c2	-INF	-90.0000	-90.0000	-24.0000

```
---- EQU shortage_capLP

              LOWER          LEVEL          UPPER         MARGINAL

h0 .c1         -INF            .            10.0000           .
h0 .c2         -INF            .            10.0000           .
h1 .c1         -INF           3.3333        10.0000           .
h1 .c2         -INF           8.9506        10.0000           .
h2 .c1         -INF            .            10.0000           .
h2 .c2         -INF            .            10.0000           .
h3 .c1         -INF            .            10.0000           .
h3 .c2         -INF            .            10.0000           .
h4 .c1         -INF            .            10.0000           .
h4 .c2         -INF            .            10.0000           .
h5 .c1         -INF            .            10.0000           .
h5 .c2         -INF            .            10.0000           .
h6 .c1         -INF            .            10.0000           .
h6 .c2         -INF            .            10.0000           .
h7 .c1         -INF            .            10.0000           .
h7 .c2         -INF            .            10.0000           .
h8 .c1         -INF            .            10.0000           .
h8 .c2         -INF            .            10.0000           .
h9 .c1         -INF            .            10.0000           .
h9 .c2         -INF            .            10.0000           .
h10.c1         -INF            .            10.0000           .
h10.c2         -INF            .            10.0000           .
h11.c1         -INF            .            10.0000           .
h11.c2         -INF            .            10.0000           .
h12.c1         -INF            .            10.0000           .
h12.c2         -INF            .            10.0000           .
h13.c1         -INF            .            10.0000           .
h13.c2         -INF            .            10.0000           .
h14.c1         -INF            .            10.0000           .
h14.c2         -INF            .            10.0000           .
h15.c1         -INF            .            10.0000           .
h15.c2         -INF            .            10.0000           .
h16.c1         -INF            .            10.0000           .
h16.c2         -INF            .            10.0000           .
h17.c1         -INF          10.0000        10.0000         -8.0000
h17.c2         -INF          10.0000        10.0000           .
h18.c1         -INF           3.3333        10.0000           .
h18.c2         -INF           6.6667        10.0000           .
h19.c1         -INF            .            10.0000           .
h19.c2         -INF           4.3827        10.0000           .
h20.c1         -INF          10.0000        10.0000        -15.0000
h20.c2         -INF            .            10.0000           .
h21.c1         -INF            .            10.0000           .
h21.c2         -INF            .            10.0000           .
h22.c1         -INF           3.3333        10.0000           .
```

```
h22.c2        -INF             .            10.0000          .
h23.c1        -INF          10.0000         10.0000       -15.0000
h23.c2        -INF             .            10.0000          .

---- EQU shortage_totalLP

          LOWER          LEVEL          UPPER         MARGINAL

c1        -INF           40.0000        40.0000       -17.0000
c2        -INF           30.0000        30.0000       -16.6667

---- EQU min_coverageLP

           LOWER          LEVEL          UPPER         MARGINAL

h0         80.0000       195.4938        +INF             .
h1         80.0000       122.7160        +INF             .
h2         80.0000       110.0000        +INF             .
h3         80.0000        83.3951        +INF             .
h4         80.0000        80.0000        +INF           56.0000
h5         80.0000        80.0000        +INF           18.0000
h6         80.0000       117.7160        +INF             .
h7         80.0000       200.0000        +INF             .
h8         80.0000       250.0000        +INF             .
h9         80.0000       310.0000        +INF             .
h10        80.0000       290.0000        +INF             .
h11        80.0000       346.6049        +INF             .
h12        80.0000       390.0000        +INF             .
h13        80.0000       412.7160        +INF             .
h14        80.0000       385.6173        +INF             .
h15        80.0000       375.6173        +INF             .
h16        80.0000       414.5062        +INF             .
h17        80.0000       420.0000        +INF             .
h18        80.0000       420.0000        +INF             .
h19        80.0000       415.6173        +INF             .
h20        80.0000       390.0000        +INF             .
h21        80.0000       360.0000        +INF             .
h22        80.0000       316.6667        +INF             .
h23        80.0000       270.0000        +INF             .

---- VAR xLP

              LOWER          LEVEL          UPPER         MARGINAL

h0 .c1          .             5.9259        +INF             .
h0 .c2          .             2.2840        +INF             .
h1 .c1          .              .            +INF            EPS
h1 .c2          .             4.3827        +INF             .
h2 .c1          .            17.2840        +INF             .
h2 .c2          .            13.3333        +INF             .
```

h3 .c1	.	0.3086	+INF	.
h3 .c2	.	12.2840	+INF	.
h4 .c1	.	.	+INF	EPS
h4 .c2	.	22.0988	+INF	.
h5 .c1	.	32.7160	+INF	.
h5 .c2	.	.	+INF	-2.13163E-14
h6 .c1	.	4.0741	+INF	.
h6 .c2	.	15.6173	+INF	.
h7 .c1	.	55.6173	+INF	.
h7 .c2	.	44.3827	+INF	.
h8 .c1	.	62.7160	+INF	.
h8 .c2	.	2.2840	+INF	.
h9 .c1	.	48.6420	+INF	.
h9 .c2	.	28.9506	+INF	.
h10.c1	.	31.8519	+INF	.
h10.c2	.	41.0494	+INF	.
h11.c1	.	43.7037	+INF	.
h11.c2	.	.	+INF	-1.42109E-14
h12.c1	.	42.0370	+INF	.
h12.c2	.	46.6667	+INF	.
h13.c1	.	.	+INF	EPS
h13.c2	.	37.7160	+INF	.
h14.c1	.	34.7531	+INF	.
h14.c2	.	8.9506	+INF	.
h15.c1	.	71.0494	+INF	.
h15.c2	.	28.9506	+INF	.
h16.c1	.	31.1111	+INF	.
h16.c2	.	34.3827	+INF	.
h17.c1	.	62.0988	+INF	.
h17.c2	.	22.2840	+INF	.
h18.c1	.	56.6667	+INF	.
h18.c2	.	43.3333	+INF	.
h19.c1	.	25.4321	+INF	.
h19.c2	.	24.3827	+INF	.
h20.c1	.	30.9877	+INF	.
h20.c2	.	.	+INF	7.105427E-15
h21.c1	.	29.3210	+INF	.
h21.c2	.	.	+INF	2.131628E-14
h22.c1	.	6.4815	+INF	.
h22.c2	.	.	+INF	2.131628E-14
h23.c1	.	.	+INF	6.0000
h23.c2	.	.	+INF	6.0000

---- VAR LLP

	LOWER	LEVEL	UPPER	MARGINAL
h0 .c1	.	.	+INF	47.0000
h0 .c2	.	.	+INF	61.6667
h1 .c1	.	3.3333	+INF	.

```
h1 .c2          .          8.9506       +INF            .
h2 .c1          .           .           +INF         23.0000
h2 .c2          .           .           +INF         20.0000
h3 .c1          .           .           +INF         47.0000
h3 .c2          .           .           +INF         50.0000
h4 .c1          .           .           +INF         47.0000
h4 .c2          .           .           +INF         50.0000
h5 .c1          .           .           +INF         47.0000
h5 .c2          .           .           +INF         61.6667
h6 .c1          .           .           +INF          3.0000
h6 .c2          .           .           +INF         61.6667
h7 .c1          .           .           +INF         21.0000
h7 .c2          .           .           +INF          5.6667
h8 .c1          .           .           +INF         68.0000
h8 .c2          .           .           +INF         49.6667
h9 .c1          .           .           +INF         36.0000
h9 .c2          .           .           +INF         47.0000
h10.c1          .           .           +INF         74.0000
h10.c2          .           .           +INF          3.0000
h11.c1          .           .           +INF         77.0000
h11.c2          .           .           +INF         47.0000
h12.c1          .           .           +INF         36.0000
h12.c2          .           .           +INF         50.0000
h13.c1          .           .           +INF         77.0000
h13.c2          .           .           +INF         23.6667
h14.c1          .           .           +INF         36.0000
h14.c2          .           .           +INF         61.6667
h15.c1          .           .           +INF         36.0000
h15.c2          .           .           +INF         61.6667
h16.c1          .           .           +INF         77.0000
h16.c2          .           .           +INF         20.6667
h17.c1          .         10.0000       +INF            .
h17.c2          .         10.0000       +INF            .
h18.c1          .          3.3333       +INF            .
h18.c2          .          6.6667       +INF            .
h19.c1          .           .           +INF         38.0000
h19.c2          .          4.3827       +INF            .
h20.c1          .         10.0000       +INF            .
h20.c2          .           .           +INF         26.0000
h21.c1          .           .           +INF         38.0000
h21.c2          .           .           +INF         55.6667
h22.c1          .          3.3333       +INF            .
h22.c2          .           .           +INF         17.6667
h23.c1          .         10.0000       +INF            .
h23.c2          .           .           +INF         37.6667

                           LOWER          LEVEL          UPPER         MARGINAL

---- VAR zLP               -INF        184811.6667       +INF             .
```

```
**** REPORT SUMMARY :        0     NONOPT
                             0 INFEASIBLE
                             0  UNBOUNDED
GAMS 52.5.0  9bc771bf Jan 28, 2026         WEX-WEI x86 64bit/MS Windows - 02/15/26
15:52:16 Page 7
G e n e r a l   A l g e b r a i c   M o d e l i n g   S y s t e m
E x e c u t i o n


----    143 PART 2 (LP relaxation) objective
            VARIABLE zLP.L                 =   184811.667


EXECUTION TIME       =        0.156 SECONDS      4 MB  52.5.0 9bc771bf WEX-WEI


USER: GAMS Demo, for EULA and demo limitations see   G251112+0003Cc-GEN
      https://www.gams.com/latest/docs/UG%5FLicense.html         DC0000


**** FILE SUMMARY

Input      C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\telecom_part2.gms
Output     C:\Users\m\OneDrive\Desktop\C238~1\NEWFOL~1\telecom_part2.lst
```